

# ♟ Blundermind – Bot Creation & Play Reference

TECHNICAL BRIEF · ENGINE, PERSONALITY, OPENINGS, TIMING, PRESSURE · V2.0

A Blundermind bot is a layered configuration that wraps a chess engine in a personality. Each layer adds constraints and biases on top of the engine's raw move candidates, producing a bot with a distinctive – and reproducible – playing style. Bots can be saved, loaded, and shared as `.json` files.

## 1 · ENGINE SELECTION

The Engine section determines *which brain* generates move candidates and *how strong* it plays. All engines share the same Centipawmeter strength dial (ELO 600-2600).

### Maia (Neural Net)

A neural network trained on tens of millions of human Lichess games. Maia produces a ranked probability distribution over legal moves rather than a single best move, which makes it uniquely suited for human-like, stylistic play. Strength is set directly in ELO – the network itself is selected to match.

ELO 600-2600

Probabilistic output

Human-pattern moves

### Stockfish (Traditional)

Stockfish 18 Lite (WASM). Alpha-beta search with NNUE evaluation. Strength is controlled via Skill Level (0-20) mapped from the ELO dial. Deterministic by default; temperature sampling can be applied on top of SF's MultiPV output to add variety.

Skill 0-20

Alpha-beta + NNUE

MultiPV ready

### Lichess-Maia

Opens from the Lichess Masters or Lichess community database for a configurable number of moves, then hands off to Maia for the midgame. The opening ELO and Maia ELO are set separately.

Explorer + Maia

### Lichess-SF

Same opening-book phase as Lichess-Maia, then falls back to Stockfish. Good for bots that should know their theory but play engine-style after the opening.

Explorer + SF

### Hybrid

Blends Maia and Stockfish by weight for every move. A 70/30 Maia/SF split plays mostly human-style but with occasional engine precision.

Weight blend

## 2 · MOVE SAMPLING – TEMPERATURE

Temperature reshapes Maia's probability distribution before sampling. It has no effect when using pure Stockfish without MultiPV.

$p'(m) = p(m)^{1/T} / \sum p(i)^{1/T}$   $T < 1$  (Focused): sharpens the distribution – top-ranked moves get even more weight  $T = 1$  (Neutral): samples Maia's unmodified distribution  $T > 1$  (Wild): flattens the distribution – lower-probability moves become reachable  $T \rightarrow \infty$  (Uniform): all legal moves equally likely  $\text{argmax}$  (Top move): always picks the single highest-probability move (deterministic)

| PRESET   | T VALUE             | BEHAVIOUR                                        |
|----------|---------------------|--------------------------------------------------|
| Top move | <code>argmax</code> | Fully deterministic – always Maia's #1 pick      |
| Focused  | <code>0.4</code>    | Mostly top 1-2 moves; small variation            |
| Neutral  | <code>1.0</code>    | Samples Maia's unmodified distribution           |
| Varied   | <code>1.8</code>    | Wider sample; surprise moves more frequent       |
| Wild     | <code>3.0</code>    | Even low-probability moves get meaningful weight |

The slider accepts any value from `0.1` to `4.0` in 0.1 increments. The live histogram shows the actual probability distribution that will be sampled from.

## 3 · PERSONALITY – THE CENTIPAWMETER & ATTRACTORS

The personality layer operates on the ranked list of candidate moves produced by the engine. It does not change what moves are *legal* – it changes which legal moves are *chosen*, subject to the centipawn budget.

### The Centipawmeter

Sets the maximum centipawn distance from the engine's best move that the personality is allowed to reach. At 0 cp the bot always plays the engine's top choice. At 200 cp it may choose any move within 200 cp of best – giving the personality wide latitude. The Move Quality Range dual-slider refines *which part* of that band is sampled.

Move Quality Range

A dual-thumb slider that selects a percentile band within the legal move list (sorted by engine score). The left thumb is the "floor" – how far down in quality the bot will reach. The right thumb is the "ceiling" – moves better than this are also excluded (useful for deliberately imprecise play). Default: 0-100% (all legal moves within the cp budget).

Personality Fingerprint (Radar)

The 11-spoke radar summarises the active attractor profile. Drag any spoke node directly, or use the sliders below the radar. The radar updates live.

4 · STRATEGIC ATTRACTORS

Each attractor adds a continuous bias score to candidate moves. Scores are summed across all active attractors; the candidate with the highest total within the cp budget is chosen. Sliders run -5 (left label) to +5 (right label); 0 = neutral (no effect).

| ATTRACTOR               | LEFT (-)              | RIGHT (+)               | MECHANISM                                                                                                                                                                                                |
|-------------------------|-----------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chaos Agent             | Simplifier            | Chaos Agent             | Scores via Stockfish MultiPV $\sigma$ (std-dev of top-N evaluations). High $\sigma$ = tactical, branching position.                                                                                      |
| Complicate When Winning | Simplify when winning | Complicate when winning | Activates when eval $\geq$ +100 cp. Boosts (or suppresses) moves that increase MultiPV variance – keeping complications alive vs. trading into a clean win.                                              |
| Space Cadet             | Space Waster          | Space Cadet             | Counts weak squares (empty, unattacked) in own camp before/after each candidate using full attack maps. Positive = reduce weak squares; negative = cede space.                                           |
| Fort Knox               | Glass Cannon          | Fort Knox               | Sums total defender count across all own pieces. Positive = maximise defensive support; negative = expose pieces aggressively.                                                                           |
| Gambito                 | Gambit Shy            | Gambito                 | Opening phase only (first 20 moves). Boosts ECO-matched gambit lines; falls back to pawn-sacrifice detection (pawn moving to opponent-attacked, undefended square).                                      |
| Trade Seeker            | Trade Avoider         | Trade Seeker            | Reweights by capture and threat value. Positive = boost captures and new threats; negative = avoid exchanges.                                                                                            |
| Structure               | Loose structure       | Rigid structure         | Pawn structure delta via Stockfish's pawn evaluation (isolated, doubled, pawn islands). Positive = consolidate; negative = open, mobile structure.                                                       |
| Hustle                  | Overthinker           | Coffeehouse Hustler     | Scales think-time allocation: $t = \text{base} \times (1 - v \times 0.15)$ . Also feeds directly into degradation curve input – a hustler at +5 thinks ~0.3 s and sees steep ELO drops.                  |
| Luck                    | Good day              | Bad day                 | Shifts the sampling window within Maia's ranked list. Positive = sample from top of distribution; negative = sample from the tail (off-form, lower-probability moves).                                   |
| Panicky                 | Calm under pressure   | Panicky                 | Scales steepness of the time-pressure ELO degradation curve ( $k$ multiplier). Positive = sharper ELO drop in the final seconds; negative = flat curve regardless of clock.                              |
| Attacker                | Peacemaker            | Attacker                | Scores by total opponent pieces under attack after the move: $\text{score} += w \times \tanh(\frac{\Sigma \text{threats}}{6})$ . Positive = maximise threat coverage; negative = prefer quiet positions. |

Piece Preferences (Pieces tab)

Six additional sliders bias move selection toward or away from moves that involve specific piece types. Range -5 to +5. Example: a Queen +4 bot will preferentially find moves where the queen participates, within the cp budget.

 **Pawn**

Boosts/suppresses pawn advances and pawn captures. Does not affect forced captures.

 **Knight**

Rewards knight manoeuvres. Combines well with closed-position preferences (rigid structure + Fort Knox).

 **Bishop**

Boosts diagonal play and bishop-pair exploitation. Works well with loose-structure preference.

 **Rook**

Rewards rook lifts and file seizure. Most impactful in the middlegame/endgame transition.

 **Queen**

Positive = active queen, early development, queen-led attacks. Negative = conservative, avoid premature queen exposure.

 **King**

Positive = king activity and endgame aggression. Negative = king-safety priority, prefer castled position.

One-Click Personalities

Pre-configured attractor profiles that set all 11 attractors at once. Any individual slider can be tweaked after applying a personality – the name clears when a manual edit is made.

| NAME                | TAG        | CHARACTER                                                                                             |
|---------------------|------------|-------------------------------------------------------------------------------------------------------|
| Captain Entropy     | Chaos      | Maximises complexity; keeps complications alive; attacks aggressively. Chaos+4, Gambit+3, Attacker+2. |
| Norm                | Order      | Prefers closed, simplified positions. Chaos-4, Complicate-4, Structure+4.                             |
| Attacky McTackerson | Captures   | Favours captures and threats within the cp ceiling. Trade+4, Attacker+3.                              |
| Overthinker         | Methodical | Deliberate; long think times; likely gets into time trouble. Hustle-4, Panicky-3, Structure+3.        |
| Coffeehouse Hustler | Fast       | Fast imprecise early play, tightens in endgame. Hustle+5, Luck+2, Gambit+2.                           |
| The Blunderer       | Human      | Plays well then cracks unexpectedly under pressure. Luck+3, Panicky+5.                                |
| The Hoarder         | Material   | Never sacrifices material voluntarily. Trade-3, Gambit-3, Luck-2.                                     |
| Pawn Chain Gang     | Pawns      | Maximises connected pawns and pawn structure. Structure+5.                                            |
| Spite Check         | Checks     | Always checks if a check exists. Chaos+3, Attacker+3, Trade+2.                                        |
| Clock Watcher       | Clock      | Conservative when ahead on time; reckless when behind. Panicky-3, Complicate-2.                       |
| Grandmaster Bad Day | Human      | Maia 2400 skill but samples from low-probability moves. Luck+5, Panicky+2.                            |

## 5 • CUSTOM CONTROLS — BUILD YOUR OWN RULE

Custom controls let you write arbitrary board-feature rules not covered by the fixed attractors. Each control selects a *metric*, optionally gates on *game phase*, *advantage state*, and *time pressure*, then applies a continuous bias to move candidates that increase (or decrease) that metric.

### Available Metrics (30+)

| CATEGORY          | METRIC                | WHAT IT MEASURES                                                    |
|-------------------|-----------------------|---------------------------------------------------------------------|
| Pawns             | Passed pawns          | Own passed pawns (no enemy pawn ahead on file or adjacent files)    |
|                   | Pawn advancement      | How far own pawns have advanced from start rank                     |
|                   | Doubled pawns         | Own doubled pawns                                                   |
|                   | Isolated pawns        | Own isolated pawns (no friendly pawn on adjacent file)              |
|                   | Pawn islands          | Number of separate pawn groups                                      |
|                   | Connected pawns       | Pawns supported by a neighbour (phalanx or chain)                   |
|                   | Pawn structure        | Overall health — fewer islands, doubled and isolated pawns          |
| Pieces & activity | Centralization        | Rewards knights, bishops, rooks and queens near the centre          |
|                   | Outposts              | Minor pieces on advanced squares defended by own pawn, unchallenged |
|                   | Mobility              | Total squares own pieces attack — proxy for activity                |
|                   | Bishop pair           | Whether own side holds both bishops                                 |
|                   | Rooks on open files   | Own rooks on files with no pawns                                    |
|                   | Rooks on the 7th      | Own rooks on the enemy's second rank                                |
|                   | Development           | Knights and bishops off back rank. Best gated to Opening phase.     |
| Material          | Material balance      | Own material minus opponent's (P=1, N/B=3, R=5, Q=9)                |
|                   | Pieces on board       | Total pieces — slide left to simplify, right to complicate          |
|                   | Queens on board       | Queens still on the board                                           |
| King              | Enemy king pressure   | Own attacks on the 9 squares around the enemy king                  |
|                   | My king safety        | Enemy attacks on the 9 squares around own king                      |
|                   | My pawn shield        | Friendly pawns sheltering own king                                  |
|                   | Gives check           | Boosts moves that put the enemy king in check                       |
| Threats           | Enemy pieces attacked | Distinct enemy pieces own side attacks after the move               |
|                   | Loose enemy pieces    | Enemy pieces attacked and undefended — free-material radar          |
|                   | My hanging pieces     | Own undefended attacked pieces                                      |
|                   | Piece defense         | Total defensive coverage of own pieces                              |
| Square control    | My weak squares       | Empty squares in own camp no longer controlled                      |
|                   | Holes in enemy camp   | Empty squares in the enemy half the opponent doesn't control        |
|                   | Central control       | Control of d4, e4, d5 and e5                                        |
|                   | Space                 | Empty squares in the enemy half own side controls                   |

### Gates — When the Rule Fires

#### Game Phase

- Any — fires every move
- Opening — first ~20 moves
- Middlegame — pieces on board
- Endgame — few pieces remain

#### Advantage State

- Any result — always active
- When winning —  $\text{eval} \geq +100$  cp
- When losing —  $\text{eval} \leq -100$  cp
- When equal —  $|\text{eval}| < 100$  cp

#### Time Pressure

- Any clock — always active
- Bot pressured — bot < 30s or <15%
- Opp pressured — opponent in time trouble
- Either side — either clock low

Custom controls can be stacked. All active controls score candidates simultaneously; the scores are summed with the attractor scores before selection.

## 6 • OPENINGS

### Off

No opening book. Engine picks from move 1.

### Main Line

Follows the most popular book moves from either the **Lichess Masters** database or the **Lichess community** database, for up to a configurable book depth (default 20 moves). A "Modern only" filter restricts to games from 2010 onward. White and Black can have *separate* settings.

### Repertoire

Define up to several opening lines by name (e.g. "Sicilian Defence", "King's Gambit"). Each line gets a percentage weight. The bot randomly selects a line each game according to the weights, then follows the Lichess book within that opening. This gives a bot a recognisable but varied repertoire. White and Black repertoires are configured independently.

Book sources: Lichess Masters Lichess Community

## 7 · MOVE TIMING

Think time is computed before the engine is queried. It combines the timing mode, the Hustle attractor, clock-pressure shortcuts, and a  $\pm 20\%$  jitter to produce an actual seconds value – which then feeds directly into the time-pressure degradation curves.

| MODE              | BEHAVIOUR                                                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed interval    | Constant pause between all moves. Set the delay in seconds. Simplest and most predictable.                                                                                            |
| Instantaneous     | Responds at engine speed (effectively 0 delay). Useful for analysis or speed variants.                                                                                                |
| Complexity scaled | Think time scales with position complexity (measured as Stockfish MultiPV variance). Tactical positions get more time; forced sequences get less. Mimics human time allocation.       |
| Mirror user       | Bot matches the human's move rhythm $\pm$ a configurable offset. If you played fast the bot plays fast; if you took 30 seconds it will too. Creates a psychologically realistic feel. |

The **Hustle attractor** (from the Personality section) acts as a global multiplier on the timing mode's base time:  $t_{\text{actual}} = t_{\text{base}} \times (1 - \text{hustle} \times 0.15)$ . A Hustle +5 bot cuts think time by 75%. This same **t\_actual** value is the input to both degradation curves – not a clock percentage.

## 8 · TIME-PRESSURE DEGRADATION

Two independent sigmoid curves describe how the bot's play deteriorates as its clock runs low. Both curves are draggable directly on the chart.

### ELO Degradation

As clock time falls below a threshold, the bot's effective ELO is reduced. The curve maps remaining seconds  $\rightarrow$  ELO adjustment (always negative or zero). Shape:  $\text{elo}(t) = E_0 - \Delta \cdot \sigma(k \cdot (\log t - 0.33))$  where  $k$  controls steepness and  $\Delta$  is max drop. The Panicky attractor multiplies  $k$ .

A bot configured as ELO 2400 in time trouble does not suddenly play like a calm 1900 – it plays like a *stressed* 2300: still strong but more likely to pick from the wrong part of the distribution.

### Move Quality Degradation

Separately, the move quality window shifts under time pressure. At full time the bot samples from its normal quality range. As time runs low, the window shifts downward (toward weaker moves) by a configurable amount. This captures the empirical finding that time-pressured players don't just play at a lower "level" – they make a specific kind of mistake: playing the first plausible move rather than the best one.

Both curves can be disabled with the Pressure Off toggle – for a bot that never cracks no matter the clock.

## 9 · SAVING, LOADING & SHARING

Any bot configuration can be saved as a **.json** file and reloaded in any Blundermind session.

### Saving a bot

- Click **Start** to lock in the current configuration.
- Use the **Save bot file** button to download a **.json** snapshot.
- Optionally give the bot a name – the name is stored in the file.

### Loading a bot

- Click **Load bot file** and select a **.json** file.
- All settings are restored: engine, ELO, temperature, all attractors, piece preferences, custom controls, openings, timing, and degradation curves.
- The loaded bot's name is displayed above the controls.

**What is saved:** engine mode, ELO, temperature value, all 11 attractor values, 6 piece-preference values, all custom controls (metric, phase, advantage gate, time-pressure gate, strength), opening mode and repertoire for each colour, booksource and depth, timing mode and parameters, both degradation curves, and personality name if selected.

Starting a game

- Configure the bot in the Bot Controls panel.
- Select your colour (White / Black / Random).
- Set the time control (untimed, or any Fischer/delay format).
- Press **Start**.
- The board switches to playing mode. Bot Controls remains open to let you monitor the bot's active state during the game.

During the game

- **Resign** – ends the game immediately; prompts for confirmation.
- **Draw offer** – bot always declines (future: configurable).
- **Save game** – download the game as PGN at any point.
- **Rematch** – immediately restarts with the same bot settings and swaps colours.
- In Expert (pro) mode: result is displayed in the notation column; Resign and Draw buttons disable after game over.

Learning Mode Indicators

While playing a bot, all board vision indicators remain available in Learning mode: threat circles, capture safety, pins, forks, discovered attacks, x-ray pressure, overloaded pieces, weak squares, and the check-threat overlay. These are independent of the bot and provide real-time positional feedback.

11 · QUICK REFERENCE

| GOAL                   | KEY SETTINGS                                                                 |
|------------------------|------------------------------------------------------------------------------|
| Human-like club player | Maia 1200-1500 · Neutral temp · Luck +2 · Panicky +1 · Fixed timing 3-8 s    |
| Tactical slugger       | Maia or SF 1800 · Varied temp · Attacker +4 · Chaos +3 · Trade +3            |
| Positional grinder     | SF 1600 · Top move or Focused temp · Structure +3 · Fort Knox +2 · Hustle -3 |
| Gambit specialist      | Maia 1400 · Varied temp · Gambito +4 · Chaos +2 · Repertoire openings        |
| Time-trouble simulator | Any engine · Pressure degradation enabled · Panicky +3 · Hustle +3           |
| Openings trainer       | Lichess-SF 2200 · Repertoire with target lines · Depth 25 · Instant timing   |
| Endgame trainer        | SF 2000 · Top move · Pawn advancement custom control (Endgame gate)          |
| Bad day Grandmaster    | Maia 2400 · Wild temp · Luck +5 · Panicky +2 · "Grandmaster Bad Day" preset  |